Double Integrals

Iterated integrals over rectangles

1 Evaluate $\int_0^2 \int_0^3 (x^2 + y)\,dy\,dx$.

2 Evaluate $\int_1^2 \int_0^1 6xy^2\,dx\,dy$.

Double integrals over general regions

3 Evaluate $\int_0^1 \int_0^x (x + y)\,dy\,dx$, integrating over the triangular region $0 \le y \le x \le 1$.

4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1$, $0 \le y \le 1$.